Casey Barton

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PROFILE

Applied AI engineer based in Honolulu. I build governed agent systems and the evaluation infrastructure that keeps them honest — optimization harnesses, offline replay, automated evaluation, synthetic task generation, regression detection, human-in-the-loop feedback, model routing and orchestration, metrics and experimentation frameworks. Measure-first delivery with explicit release thresholds is my operating standard, not a phase.

Three years running an evidence-grade agent control plane: typed state transitions, approval gates, audit receipts, recoverable runs, structured observability, failure analysis with root-cause quarantine instead of silent retries. Reports to Kyle Maxwell's AI developer platform team — the domain is the same: measuring and improving agents' impact on engineering velocity.

CORE COMPETENCIES

Agent Evaluation & Optimization: evaluation harnesses · automated evaluation · offline replay · synthetic task generation · regression detection & gating · LLM-as-judge · gold sets & release thresholds · metrics & experimentation frameworks · model routing & orchestration (quality / latency / cost) · agent behavior analysis & failure investigation

AI & Agents: hands-on AI agent development & deployment (tool/API use, orchestration, state, failure handling) · MCP protocol · LLM retrieval & structured outputs · human-in-the-loop controls · multi-agent orchestration

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · Supabase · Postgres · Vercel · GitHub Actions (CI/CD) · large-scale production software with cross-team integration

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Evaluation-first delivery (job-app-helix): **989-test** agent control plane; evals with explicit release thresholds, gold sets, and LLM-as-judge; evidence state machine (nominal / recoverable / hard-no-go) with the hard-failure path exiting non-zero and CI asserting it — offline regression gating before anything promotes.
- Optimization-harness mechanics (anthropic-agent-coordinator): deterministic DAG scheduler, **62/62** tests green; agent-behavior tooling to investigate failures and accelerate iteration; tool calls proposed for review, never executed blind — human-in-the-loop control built into the orchestration layer.
- Model-agnostic orchestration (unified-memory-mcp, mcp-iceberg-mansion): Mem0, Supermemory, Supabase pgvector, Pinecone, and Neo4j federated behind one MCP interface — routing across models and providers with provenance and structured outputs; parallel fan-out with content-hash dedup.
- Production feedback loops (apex-control-plane): cloud deployment, access controls, SHA-256 transport envelopes, idempotency keys, circuit breakers, dead-letter capture, fail-closed startup;

failure analysis with root-cause quarantine instead of silent retries — production usage feeding back into iteration.

- Operate a **1,398-repository** applied-AI estate (live-verified 2026-10-20: 695 public + 703 private) — **65 MCP connectors** integrated with claim tracking and idempotent automation; reusable agents, skills, connectors, and playbooks compiled as the compounding output; **20+ production repositories** maintained under test and CI.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program (brief attendance)

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — regression-gated agent control plane

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions — automated evaluation with real veto power, review gates that gate.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass; regression detection as an honesty guarantee, not a checkbox.

Anthropic-agent-coordinator — orchestration with built-in evaluation hooks

- Deterministic DAG scheduling with review-gated tool calls (**62/62** tests green) — the mechanics an agent optimization harness instruments: behavior analysis, failure investigation, iteration speed.
- Metrics and run-state observability structured by design — typed states and receipts that an experimentation framework can measure.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

Evaluation: evaluation harnesses · automated evaluation · offline replay & synthetic tasks · regression detection & gating · LLM-as-judge · gold sets · release thresholds · metrics & experimentation frameworks

AI / Agents: agents & multi-agent orchestration · MCP tool integration patterns · RAG · model routing & orchestration · prompt & context engineering · human-in-the-loop controls

Data & Infra: Supabase · Postgres · Vercel · GitHub Actions · REST APIs · observability & failure analysis · governance & reversibility

Evidence boundary: portfolio repositories are independent work. Evaluation depth comes from my own agent systems estate — no Reddit-platform production depth claimed, no claim of having shipped Reddit-internal platforms. Domain-inspired demonstrations are not company affiliation or certification.